6(a) (i) When the slide contact S is moved all the way up to the top of the 20 kΩ resistance, $V_{xs} = 0$ V.

When the slide contact S is moved all the way down, V_{xs} can be found using the potential divider principle:

$$V_{xs} = \left(\frac{20}{20+4}\right)12 = 10\text{V}$$ M1

Hence, the voltage range of V_{XS} is <u>0 V to 10 V</u>. A1

<u>Comments</u>
The key concept is potential divider in accordance to ratio of the resistances. Some candidates could not identify 0V value when there is 0 Ω resistance.

(ii) When the slide contact S is moved all the way up to the top of the 20 kΩ resistance,

$V_{sz} = 12 – 0 = 12$ V

When the slide contact S is moved all the way down,

$V_{sz} = 12 – 10 = 2$V

Hence the voltage range of V_{SZ} is <u>2 V to 12 V</u>. A1

<u>Comments</u>
Alternative method. By looking at the answer in part (i). When V_{XS} is 0 V, V_{SZ} is 12 V. When V_{XS} is 10 V, V_{SZ} is 2 V.

(iii) When the slide contact is placed at midpoint of the 20 kΩ track, R = 10 kΩ

$$\frac{1}{R_{SY}} = \frac{1}{10} + \frac{1}{10}$$ [1]

$$R_{SY} = 5\ \text{k}\Omega$$

Using potential divider principle,

[1]

$$V_{SY} = \frac{5}{10+5+4}(12) \approx 3.2\ \text{V}$$

The voltmeter reading = V_{SY}
≈ 3.2 V

Comments
Alternative method. By looking at the answer in part (i). When V_{XS} is 0 V, V_{SZ} is 12 V. When V_{XS} is 10 V, V_{SZ} is 2 V.

(b) **(i)** Since the voltmeter reading is 3.0 V when the switch is opened, the e.m.f. of the battery is 3.0 V.

When switch is closed,
p.d. across siren and warning lights = 1.7 V
Total current through siren & warning lights = 1.7 / 1.5 + 1.7 / 2.4 + 1.7/ 2.4
= 2.55 A [C1]

p.d. across internal resistance = 3.0 - 1.7 = 1.3 V
internal resistance = 1.3 / 2.55 [M1]
= 0.51 Ω [A1]

Comments
A common misconception is to consider the internal resistance as parallel to the siren and warning lights. Students should note that the siren and the warning lights are in parallel. The internal resistance is in series with the combined arrangement of siren and warning lights.

(ii) 1. The p.d. across the siren (or current through siren) decreases and hence less loud [B1].
2. The p.d across the warning lights (or current through lights) decreases and hence dimmer lights. [B1]

Comment
This part was well-attempted.

7 **(a)** Current I pointing downwards. A1